Cohomology rings of higher Verlinde categories

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Abstract

We prove the Benson–Etingof conjecture describing the unit cohomology rings of $\mathsf{Ver}_{p^{m+1}}$ over algebraically closed fields of every positive characteristic. The isomorphisms preserve the standard parameters and are compatible with the inclusions of the Verlinde tower. The proof combines exact transport of rank-one costandard sequences with an idempotent corner whose algebra is a tensor product with a Brauer line, or with the dual numbers in characteristic two. We identify every coefficient comparison map by a truncated Yoneda extension. This determines the full image in the localized ring, including its nilpotent elements, and permits exact descent of multiplication.

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1 Statement and organization of the proof

Let $\mathbf{k}$ be an algebraically closed field of characteristic $p > 0$. Write

$$\mathcal{C}_r = \mathsf{Ver}_{p^r}, \qquad H_r = \mathrm{Ext}^{\bullet}_{\mathcal{C}_r}(\mathbf{1}, \mathbf{1}), \qquad H_1 = \mathbf{k}.$$

The standard inclusions $\mathcal{C}_r \hookrightarrow \mathcal{C}_{r+1}$ induce homomorphisms $H_r \to H_{r+1}$. All products of extensions below are Yoneda products, and tensor products of graded algebras use the Koszul sign rule. For a rationally graded algebra, Int denotes the subalgebra consisting of its integer-degree homogeneous components. Set

$$\mathcal{E}_m(2) = \mathsf{Int}(\mathbf{k}[x_1, \dots, x_m]), \qquad |x_i| = 1 - 2^{-i}. \tag{1.1}$$

$$\begin{aligned} \mathcal{E}_m(p) = \mathsf{Int}(\mathbf{k}[x_1, \dots, x_m] \otimes \textstyle\bigwedge(\xi_1, \dots, \xi_m)) \qquad (p > 2), \\ |x_i| = 2 - 2p^{-i}, \qquad |\xi_i| = 1 - 2p^{-i}. \end{aligned} \tag{1.2}$$

Here x_i is even and ξ_i is odd when $p > 2$. These are the algebras of Benson–Etingof [2, §3]. Put $\mathcal{E}_0(p) = \mathbf{k}$.

Theorem 1.1. *For every prime p and every $m \geq 0$ there are graded algebra isomorphisms*

$$H_{m+1} \cong \mathcal{E}_m(p)$$

compatible with all standard inclusions. They send the actual standard parameter y_i to $x_i^{p^i}$, with

$$|y_i| = \begin{cases} 2^i - 1, & p = 2, \\ 2(p^i - 1), & p > 2. \end{cases}$$

For $m \geq 1$, they also preserve the last-step internal grading: x_m has weight one if $p = 2$, and weight two if $p > 2$; all older variables, and ξ_m when present, have weight zero. The ring is free over $S_m = \mathbf{k}[y_1, \dots, y_m]$, of rank

$$\begin{cases} 2^{m(m-1)/2}, & p = 2, \\ 2^m p^{m(m-1)/2}, & p > 2. \end{cases}$$

The categories in the theorem were constructed in characteristic two by Benson–Etingof [1], and in all positive characteristics by Benson–Etingof–Ostrik [3] and, independently, Coulembier [4]. The theorem addresses the cohomology-ring conjecture [2, Conjecture 1.3]. The parameters in the statement are the classes of the standard relative Koszul periods; their choices of nonzero scalar do not affect the assertion. At each induction step a polynomial generator in the corner is normalized to retain the chosen parameter.

The main point is a comparison of maps. An isomorphism of graded parameter modules would not determine multiplication. We instead identify the actual image of cohomology in a common localized algebra. The proof proceeds in the following order.

(1) Rank-one costandard sequences give a relative period in the old category. Its connecting maps identify every coefficient group needed at the next level with an internal-weight tail of the old ring; see Section 3.

(2) These tail identifications prove global parameter freeness at the new level, before localization is used. In particular the localization map needed for descent is injective.

(3) The restriction of the new relative period to a regular corner is determined on both terms and maps. Its complex is a specified Yoneda splice of old short extensions and endpoint extensions of a Brauer line, or of the dual numbers.

(4) Comparing the two hypercohomology spectral sequences gives the localized ring. A prefix of the restricted period, composed with a socle map, then determines each actual coefficient image. This is the comparison proved in Section 7.

(5) Those images are exactly the monomial tails in the model. Equality of their submodules inside the localized algebra is therefore equality of their subalgebras, proving global descent.

The induction uses the full ring assertion only at the old level. In particular global freeness and the new coefficient comparison are proved before the new multiplication is identified.

In a typical induction step, H_r is the known ring and H_{r+1} is the ring to be determined. The class $t = y_{r-1} \in H_r$ is inverted; the new parameter is y_r. The old relative period has coefficients in $\mathcal{C}_{r-1}$, whereas the new one has coefficients in $\mathcal{C}_r$. We keep these two relative gradings distinct throughout. In the corner, B_r always denotes the old principal-block algebra; the auxiliary algebra Z or Z_{p-1} records the new endpoint direction.

2 Categorical conventions and free modules

We record the categorical input, including the identifications on morphisms that will be used. The indecomposable tiltings for SL_2 are denoted T_i, and the same symbol denotes their images when the category is specified. Let V_r be the image of the natural module in $\mathcal{C}_r$; put $V_1 = 0$ if $p = 2$. Simple labels are the surviving ordinary highest weights $0 \le i < (p-1)p^{r-1}$; in particular $J_r(L_i) = L_{pi}$. The natural parity is the highest-weight label modulo two. There are no morphisms or extensions between the two parity components. All simple objects and all indecomposable projectives of $\mathcal{C}_r$ are self-dual. The latter are images of the self-dual tiltings with labels $p^{r-1} - 1 \le i \le p^r - 2$. We write $\mathcal{C}_r^0$ for the principal block, and B_r for its basic algebra, with left-module convention $B_r = \mathrm{End}(P_r)^{\mathrm{op}}$ for a basic projective generator. These descriptions of simples and projectives are part of [3, §4.1, §4.11]; their native self-duality also follows from the corresponding self-duality of the ordinary simple and tilting SL_2-modules.

We use the following established descriptions.

(1) The standard inclusion $J_r : \mathcal{C}_{r-1} \hookrightarrow \mathcal{C}_r$ is induced by Frobenius twist on the expanded tilting category $\overline{\mathcal{T}}_{r-1}$, whose objects X satisfy that $\mathrm{St}_{r-2} \otimes X$ is tilting; here $\mathrm{St}_j = T_{p^j-1}$. The expanded category is rigid and monoidal. Bounded exact sequences with terms in $\overline{\mathcal{T}}_r$ remain exact in $\mathcal{C}_r$; costandards $\nabla(i)$ with $0 \le i \le p^r - 2$ map to $\mathrm{Sym}^i V_r$. These are [5, Propositions 4.2, 4.8, 4.9 and Remark 4.10].

(2) Let Λ_r be the three-term exterior algebra of the natural tilting, transported to $\mathcal{C}_r$. Its components are $\mathbf{1}, V_r, \mathbf{1}$ in degrees $0, 1, 2$. For $r \ge 2$ it is the exterior algebra used in [2, §6]. At the first step one uses its transported three-term version: $\Lambda_1 = \mathbf{k}[\epsilon]/(\epsilon^2)$ with $|\epsilon|_{\mathrm{int}} = 2$ for $p = 2$, and the truncated exterior algebra of the odd line for $p = 3$. This is the exception described in [2, Remark 6.9]. There is an abelian equivalence

$$F_r : \mathcal{C}_r^0 \xrightarrow{\sim} (\mathcal{C}_{r-1})_{\Lambda_{r-1}} \qquad (r \ge 2) \tag{2.1}$$

from [3, Proposition 4.57]. Forgetting the module action is exact. Free modules are also coinduced, since Λ_{r-1} is Frobenius with top component $\mathbf{1}$ in degree two.

(3) If $D = p^r - 2$ and $X_i = \mathrm{Sym}^i V_r$, then $X_D = \psi_r$, where ψ_r is the odd invertible object for $p > 2$ and is the unit for $p = 2$. There are perfect pairings

$$X_i^* \cong X_{D-i} \otimes \psi_r. \tag{2.2}$$

The natural adjacent Koszul sequences are exact up to the top; when $D \geq 1$, the last map $X_{D-1} \to V_r \otimes X_D$ is an isomorphism. For $p = 2, r = 1$ one has $D = 0$, $V_1 = 0$, and only $X_0 = \mathbf{1}$. See [2, Proposition 6.1 and Corollaries 6.4, 6.7].

Only abelian equivalence is asserted in (2.1). Native duals and native tensor products always refer to $\mathcal{C}_r$. All ring maps and coefficient actions in the proof are maps of Yoneda extensions induced by exact functors or by precomposition. The projective-generator normalization of F_r in [3, Proposition 4.57] is used throughout.

2.1 Internal weights and Yoneda conventions

A graded object is a direct sum of objects of $\mathcal{C}_{r-1}$ indexed by integers. For a graded module M let $M\{u\}$ denote the shift with $(M\{u\})_j = M_{j-u}$. We use

$$\mathrm{Ext}^{n,w}(M, N) = \mathrm{Ext}^n_{\mathrm{gr}}(M, N\{w\})$$

for its internal-weight-w extensions. Thus a socle in module degree $-w$ contributes a class of weight w from the unit. All graded module homomorphisms used to construct the coresolutions have degree zero. Forgetting the grading gives the direct sum of these weight spaces in each Ext group.

The weights come from scaling Λ_{r-1}; they are independent of cohomological degree. They are not a grading on the native tensor product of the principal block. For coefficients we use the right action of unit cohomology by precomposition. A connecting extension acts by postcomposition and is therefore right-linear, without an additional sign. The only interchange signs used below are those in external Yoneda products over tensor-product algebras.

2.2 Native descriptions of free and inflated modules

For $Y \in \mathcal{C}_{r-1}$ of parity ε define

$$D_r(Y) = \begin{cases} J_r(Y) \otimes T_{2p-2}, & p > 2,\ \varepsilon = 0, \\ J_r(Y) \otimes T_p, & p > 2,\ \varepsilon = 1, \\ J_r(Y) \otimes T_2, & p = 2. \end{cases} \tag{2.3}$$

Lemma 2.1. *There are natural identifications of exact functors*

$$F_r D_r(Y) = Y \otimes \Lambda_{r-1}, \qquad D_r(Y)^* = D_r(Y^*).$$

For Y in the old principal block, $F_r J_r(Y)$ is the augmentation module with underlying object Y. Consequently old cohomology acts under derived coinduction by its usual right Yoneda action on the old coefficient group.

Proof. For an old projective $Y = T_i$, Donkin's tilting tensor product formula [3, Proposition 3.1(iii)] gives

$$D_r(T_i) = T_{a_i + pi}, \qquad a_i = \begin{cases} 2p-2, & i \text{ even}, \\ p, & i \text{ odd}, \end{cases}$$

when $p > 2$; for $p = 2$ one has $a_i = 2$. These are precisely the projective generators used in the proof of [3, Proposition 4.57]. Their morphisms are identified there with $\mathrm{Hom}(T_i, T_j \otimes \Lambda_{r-1})$, with composition induced by multiplication in Λ_{r-1}. A morphism induced by $J_r(f) \otimes 1$ is $f \otimes 1$ under this calculation: the identity of the restricted tilting is the degree-zero unit. Thus the free-module identification holds on the generators and all their morphisms. Both functors

are exact on each parity component, so projective presentations extend it naturally to every Y. The dual assertion follows from self-duality of the restricted tiltings and the native tensor inclusion J_r.

For the augmentation assertion, first take an old projective target $Y = P_j$. In Newton's realization [5, Theorem 5.12], Hom out of a new projective is computed by the ordinary Hom before the Serre quotient. This follows from the projective-generator description in that theorem, or its Lemma 5.6(1). Here $D_r(P_i)$ is such a new projective, while $P_j^{(1)}$ has highest weight at most $p(p^{r-1}-2) < p^r - 1$. We may therefore use the ordinary Frobenius-kernel calculation with target $P_j^{(1)}$. Both $\mathrm{Hom}(D_r(P_i), J_r(-))$ and $\mathrm{Hom}(P_i, -)$ are exact: their source objects are projective and J_r is exact. The calculation consequently extends naturally from projective presentations to every old principal-block target Y, and gives

$$\mathrm{Hom}_{\mathcal{C}_r}(D_r(P_i), J_r(Y)) = \mathrm{Hom}_{\mathcal{C}_{r-1}}(P_i, Y).$$

The restricted tilting T_{2p-2} has a one-dimensional trivial head over the first Frobenius kernel; its radical acts as zero on that head. For $p > 2$, T_p has no trivial head, and both sides of the display are zero for odd P_i when Y is in the even principal block. The degree-zero algebra acts on the displayed spaces in the usual old way and the positive ideal acts by zero. They therefore represent exactly the augmentation module Y. This calculation is natural on projective presentations; for $p = 2$ it uses $T_p = T_{2p-2} = T_2$ and the display holds for both old parities. It identifies the standard inclusion with inflation on the old principal block. Finally derived coinduction adjunction commutes with precomposition by an inflated old extension, which proves the last assertion. □

Lemma 2.2 (Graded coinduction)**.** *Place an old coefficient Y in degree zero and give the free module $Y \otimes \Lambda_{r-1}$ generator degree g. Then*

$$\mathrm{Ext}^{n,w}_{(\mathcal{C}_{r-1})_{\Lambda_{r-1}}}(\mathbf{1}, Y \otimes \Lambda_{r-1}\{g\}) = \begin{cases} \mathrm{Ext}^n_{\mathcal{C}_{r-1}}(\mathbf{1}, Y), & w = -g-2, \\ 0, & w \neq -g-2. \end{cases}$$

The identification commutes with connecting maps and with precomposition by inflated old extensions.

Proof. The multiplication pairing of Λ_{r-1}, followed by projection to its degree-two unit, is nondegenerate. It identifies its regular right module with its coinduced unit, shifted by two. Thus the displayed free module is the coinduction of $Y\{g+2\}$. The forgetful functor is exact; its right adjoint coinduction preserves injectives and, here, is itself exact because it is a shift of the free functor. Derived adjunction consequently reduces the displayed group to old graded Ext from $\mathbf{1}$ to $Y\{g+2+w\}$. Distinct internal degrees in the graded old category have no extensions. Only $g+2+w = 0$ remains, proving the formula. Naturality of the exact adjunction gives the last assertions. This argument also applies to both exceptional first algebras Λ_1. □

Lemma 2.3 (Bricks and adjacent maps)**.** *For $0 \le i \le D = p^r - 2$ one has $\mathrm{End}(\mathrm{Sym}^i V_r) = \mathbf{k}$. For $0 \le i < D$,*

$$\dim \mathrm{Hom}(V_r \otimes X_i, X_{i+1}) = \dim \mathrm{Hom}(X_i, V_r \otimes X_{i+1}) = 1.$$

Proof. Put $b = (p-1)p^{r-1}$. Newton's abelian description is the Serre quotient of the ordinary modules with weights less than $p^r - 1$ by the simples with labels $b, \dots, p^r - 2$ [5, Theorem 5.12]. On objects whose factors all have labels less than b, this quotient is fully faithful. Indeed, in the Serre-quotient Hom formula no nonzero target subobject or source quotient can belong

to the deleted subcategory. For $i < b$, this applies to $\nabla(i)$, whose endomorphisms are scalars. For $i \geq b$, apply (2.2); $D - i < b$, giving the same result.

If $i + 1 < b$, apply $\mathrm{Hom}(-, \nabla(i+1))$ to

$$0 \to \nabla(i-1) \to V \otimes \nabla(i) \to \nabla(i+1) \to 0.$$

With $\nabla(-1) = 0$, the Hom from $\nabla(i-1)$ to $\nabla(i+1)$ is zero: the latter has essential simple socle $L(i+1)$, absent from the source. The preceding full-faithfulness calculation therefore gives the first asserted dimension. If $i + 1 \geq b$, duality reduces it, with $h = D - i < b$, to $\mathrm{Hom}(X_{h-1}, V_r \otimes X_h)$. Rigid adjunction reduces this to the low-weight case. Adjunction also gives the second asserted dimension. For $p = 2, r = 1$ there is only $X_0 = \mathbf{1}$, and no adjacent claim. □

3 Relative periods and coefficient tails

Fix $r \geq 2$ and set

$$q = p^{r-1}, \qquad a = q - 1, \qquad t = y_{r-1},$$
$$d = \begin{cases} a, & p = 2, \\ 2a, & p > 2, \end{cases} \qquad T = \begin{cases} q, & p = 2, \\ 2q, & p > 2. \end{cases}$$

Let $Y_k = \mathrm{Sym}^k V_{r-1}$, with $Y_{-1} = Y_a = 0$, and put

$$M_k = \mathrm{Sym}^{s_k} V_r, \qquad s_k = \begin{cases} 2k, & p = 2, \\ pk, & p > 2,\ k \text{ even}, \\ p(k+1) - 2, & p > 2,\ k \text{ odd}. \end{cases}$$

Lemma 3.1. *There are exact sequences*

$$0 \to M_k \xrightarrow{\iota_k} Q_k \xrightarrow{\pi_k} M_{k+1} \to 0, \qquad Q_k = D_r(Y_k), \quad 0 \leq k < a. \tag{3.1}$$

Under the forgetful functor UF_r,

$$UF_r(M_k) = Y_{k-1} \oplus Y_k. \tag{3.2}$$

The endpoint M_a is the simple augmentation module ψ_{r-1}, and

$$\mathrm{Hom}(M_k, Q_k) = \mathbf{k}\iota_k, \qquad \mathrm{Hom}(Q_k, M_{k+1}) = \mathbf{k}\pi_k. \tag{3.3}$$

Proof. Dualizing the ordinary Weyl-module sequence [6, (2), §3] gives

$$0 \to \nabla(p(j-1) + p - 2 - i) \to \nabla(j-1)^{(1)} \otimes T(p+i) \to \nabla(pj + i) \to 0.$$

For odd p, take $i = p - 2$ for odd j and $i = 0$ for even j. For $p = 2$, take $i = 0$ throughout. With $j = k + 1$ the endpoints are $\nabla(s_k), \nabla(s_{k+1})$ and the middle term is the ordinary version of $D_r(Y_k)$. All weights are at most $p^r - 2$. The costandards belong to $\overline{\mathcal{T}}_r$; Frobenius twist and tensoring with the indicated tilting put the middle term there as well. Exact transport and Lemma 2.1 give (3.1), with its actual maps.

The maps between successive free terms have exterior degree one. Indeed their coefficients have opposite old parity, excluding exterior degrees zero and two in free adjunction. Start the initial unit in degree zero; the free generator of Q_k then has degree $-k-2$. Inductively M_k has Y_{k-1}, Y_k in degrees $-k-1, -k$. Its component into the top Y_k of Q_k is a monic endomorphism,

hence an isomorphism. The other component is the nonzero map $Y_{k-1} \to V_{r-1} \otimes Y_k$. Lemma 2.3 and the adjacent Koszul sequence identify its cokernel with Y_{k+1}. At the last step the top pairing makes this map an isomorphism and $Y_a = 0$. This proves (3.2). The endpoint has only $Y_{a-1} = \psi_{r-1}$ in degree $-q$, so its augmentation ideal acts by zero. At $p = 2, r = 2$ the argument is simply $0 \to \mathbf{1} \to \mathbf{k}[\epsilon]/(\epsilon^2) \to \mathbf{1} \to 0$.

Free/coinduced adjunction identifies the Hom spaces in (3.3) with

$$\mathrm{Hom}(Y_{k-1} \oplus Y_k, Y_k), \qquad \mathrm{Hom}(Y_k, Y_k \oplus Y_{k+1}).$$

The adjacent summands have opposite parity; the remaining endomorphism space is $\mathbf{k}$ by Lemma 2.3. □

For $p > 2$, concatenate the antiperiod (3.1) with its native dual. Its endpoint M_a is self-dual because it is simple. This gives a period with free terms

$$Q_0, \dots, Q_{a-1}, Q^*_{a-1}, \dots, Q^*_0. \tag{3.4}$$

For $p = 2$, $M_a = \mathbf{1}$ and the first antiperiod is already a period, with just $Q_0, \dots, Q_{a-1}$. Repeat the appropriate period and denote its cycles by C_j. Thus

$$C_j = M_j \ (0 \le j \le a), \qquad C_{d-j} = M^*_j \ (0 \le j \le a) \quad (p > 2), \qquad C_d = \mathbf{1}. \tag{3.5}$$

Lemma 3.2 (Purity, the actual period, and tails)**.** *The repeated coresolution has a graded lift whose column i in the hypercohomology spectral sequence is pure of weight*

$$f(i) = i + \lfloor i/a \rfloor.$$

Its period is a nonzero scalar multiple of the standard class t, of bidegree (d, T), and its periodic shift is cup multiplication by that class. Normalize one splice accordingly. For $0 \le j \le d$, let $I_b = \bigoplus_{w \ge b} (H_r)_{\mathrm{int}=w}$. The connecting map of the first j terms is an injective right H_r-linear map

$$\delta_j : \mathrm{Ext}^e_{\mathcal{C}_r}(\mathbf{1}, C_j) \xrightarrow{\sim} (I_{f(j)})^{e+j}, \qquad e \ge 0. \tag{3.6}$$

Every free $D_r(Y)$ satisfies $t\,\mathrm{Ext}^\bullet(\mathbf{1}, D_r(Y)) = 0$.

Proof. The first free-generator degrees were computed above as $-i - 2$. When $p > 2$, the second half has coefficients $Y^*_{a-1-i} = Y_i \otimes \psi_{r-1}$ by (2.2). Its first generator has degree $-q - 2$ and subsequent ones $-q - i - 2$. Each splice factors through an augmentation simple and its socle inclusion in a free module, so it has exterior degree two. All other maps have exterior degree one by parity. For $p = 2$ there is one such splice per period; for odd p there are two. The shifts are therefore T.

Lemma 2.2 gives the column weights directly. The generators $-i - 2$ in the first half give weight i; the generators $-q - i - 2$ in the second half give weight $q + i$. Subsequent periods add T. In column order these are exactly $f(i) = i + \lfloor i/a \rfloor$, which is strictly increasing. Thus

$$E_1^{i,b} = \mathrm{Ext}^b(\mathbf{1}, Q^i) \Longrightarrow H_r^{i+b} \tag{3.7}$$

has no differentials, and each internal weight has just one column. The class of one full period is the connecting image of the underlying identification $\mathbf{1} \to C_d = \mathbf{1}\{-T\}$, which has internal weight T. In weight T every preceding column is zero. The long exact sequence for the first d terms therefore identifies that one-dimensional Hom space with the bidegree-(d, T) component of H_r. In particular the period class is nonzero. The first free terms of the standard Koszul antiperiod are precisely $D_r(Y_0), \dots, D_r(Y_{a-1})$. Free adjunction, parity and Lemma 2.3 make each successive map space one-dimensional. Its nonzero maps therefore agree with these

maps after scalar rescaling. The endpoint maps are the unique augmentation quotient and socle inclusion. For odd p, the perfect pairings (2.2) similarly identify its second half with the native dual half. Thus this is the standard Koszul period up to scalar. Its first endpoint is in degree $-q$, from the free-generator degrees $-i-2$; its full internal shift is T, also agreeing with the total degree of top Koszul homology in [2, Corollary 6.6]. For $p=2, r=2$ the standard period is directly the one-step socle sequence of the dual numbers, of bidegree $(1,2)$. The period classes consequently differ by a nonzero scalar. Yoneda composition with the period is the periodic shift on the coresolution and hence on its coefficient columns.

For the tail assertion truncate the first j terms to P_j. There is a triangle

$$\mathbf{1} \to P_j \to C_j[-j+1] \to \mathbf{1}[1].$$

The derived Hom of P_j consists exactly of the weights $f(0), \dots, f(j-1)$ in (3.7). Projection from the full coresolution is the identity on these columns. Consequently the map

$$H_r^n \longrightarrow \mathrm{H}^n \mathbf{R}\operatorname{Hom}(\mathbf{1}, P_j)$$

is surjective in every degree and an isomorphism on each retained weight. Surjectivity in degree $e+j-1$ kills the term preceding the connecting map in the long exact sequence, so that map is injective. Its image is the kernel of projection in degree $e+j$, namely the weights at least $f(j)$. Their columns have indices at least j, which also verifies the assertion for all $e \geq 0$ without a negative-degree correction. The connecting map is composition with the extension $C_j \to \mathbf{1}[j]$; it commutes with right Yoneda precomposition, proving its linearity.

Finally Lemma 2.2 makes extensions to an unshifted free module pure of weight -2. Precomposition with t shifts that weight by $T \neq 0$, so its action on this group is zero. The same reasoning applies after any grading shift of that free module. □

Remark 3.3. Internal weight refers to scaling the graded algebra Λ. It is distinct from cohomological degree: the old period has degree d and weight T. In particular weight $q-1$ is absent from H_r, since it is skipped by f. These statements include the exceptional first algebra Λ_1 specified above.

4 Freeness before localization

We describe an induction step under the hypothesis that $H_r = \mathcal{E}_{r-1}(p)$, preserving the standard parameters and the last internal grading. All comparisons with old model monomials in this section refer to that hypothesis.

The relative Koszul antiperiod in $\mathcal{C}^0_{r+1}$ has free coefficients $X_i = \operatorname{Sym}^i V_r$, $0 \leq i \leq p^r - 2$. Its first free-generator degrees are $-i-2$. For odd p, take its native dual for the second half; its coefficients, in reverse order, are X_i^*. The perfect pairings (2.2) identify this with the usual second Koszul family. Adjacent free-module maps have one-dimensional Hom spaces by Lemma 2.3, so the comparison retains the actual differentials up to scalar. The resulting full period is the actual y_r up to scalar. The purity argument of Lemma 3.2 applies at this level as well.

For odd p, linkage puts precisely $X_{s_k} = M_k$, $0 \leq k \leq a$, in the old principal block. For $p=2$ precisely $X_{2k} = M_k$ are in it. The collapsed spectral sequence gives a decomposition over the actual parameter action:

$$H_{r+1} = \mathbf{k}[y_r] \otimes \begin{cases} \bigoplus_{k=0}^{a} \operatorname{Ext}^\bullet(\mathbf{1}, M_k)\langle 2k \rangle, & p = 2, \\ \bigoplus_{k=0}^{a} \left(\operatorname{Ext}^\bullet(\mathbf{1}, M_k)\langle s_k \rangle \oplus \operatorname{Ext}^\bullet(\mathbf{1}, M_k^*)\langle 2p^r - 3 - s_k \rangle \right), & p > 2. \end{cases} \tag{4.1}$$

Here $\langle b\rangle$ raises total cohomological degree by b. The new weights are $2k$ for $p = 2$; for odd p they are s_k and $2p^r - 2 - s_k$, respectively. This is the unit case of [2, Corollaries 7.2, 7.20]; the preceding purity calculation specifies its internal grading. Multiplication by y_r is the periodic shift. Old parameter actions agree with coinduction by Lemma 2.1.

Proposition 4.1. *Under the induction hypothesis, H_{r+1} is free over $S_r = \mathbf{k}[y_1, \dots, y_r]$. In particular its localization at $t = y_{r-1}$ is injective.*

Proof. The model has an S_{r-1}-basis consisting of its integer-degree monomials with polynomial exponent of x_i between zero and $p^i - 1$, retaining all exterior choices that meet the integer-degree condition. Reducing exponents modulo p^i gives a unique expression in this basis. The basis is homogeneous in the old last internal weight, and its weights lie in $[0, T)$. Multiplication by t adds T to that weight. If u_ν are these basis vectors and w_ν their weights, then for each $0 \le b \le T$ the tail has the explicit basis

$$\{t^{\epsilon_\nu} u_\nu\}_\nu, \qquad \epsilon_\nu = \begin{cases} 0, & w_\nu \ge b, \\ 1, & w_\nu < b. \end{cases}$$

Indeed an expansion in the old parameter basis is unique; a coefficient of u_ν lies in the tail precisely when its t-power is at least ϵ_ν. All older parameters have internal weight zero. Thus the tail is free over the actual S_{r-1}. Equations (3.5) and (3.6) identify all coefficient groups in (4.1) with such tails, with their cohomological shifts. This proves freeness over the actual S_r and the claimed injectivity. □

5 The regular corner

5.1 Socles in a corner

Recall that a basic algebra is weakly symmetric if it is self-injective and each indecomposable projective has its own simple head as its simple socle. The basic algebras presenting $\mathcal{C}_r$ have this property: native duality takes projectives to injectives and fixes both the projectives and their simple heads. The property holds on either module side.

Lemma 5.1. *Every idempotent corner of a finite-dimensional basic weakly symmetric algebra is weakly symmetric.*

Proof. Use right modules and write $1 = \sum_i e_i$ for primitive orthogonal idempotents. The socle of $e_i A$ is a line $\mathbf{k}\ell_i \subset e_i A e_i$. Choose a functional $\lambda : A \to \mathbf{k}$ which vanishes on all off-diagonal spaces $e_i A e_j$, $i \ne j$, and is nonzero on each ℓ_i. Every nonzero right ideal of A contains some ℓ_i: its intersection with the socle of the regular module is nonzero, and that socle contains each simple exactly once. It follows that $\ker \lambda$ contains no nonzero right ideal. Hence $(x, y) \mapsto \lambda(xy)$ is nondegenerate.

It suffices, after conjugation, to take e to be a sum of some e_i. If $0 \ne x \in eAe$, the nonzero right ideal $xA \subset eA$ contains a line $\mathbf{k}\ell_i$ with $e_i \le e$. Choose a with $xa = \ell_i$. Then $x(eae) = \ell_i$, so the restriction of the multiplication pairing to eAe is also nondegenerate. Thus eAe is Frobenius, in particular self-injective. Furthermore $\operatorname{rad}(eAe) = e \operatorname{rad}(A) e$: the ideal on the right is nilpotent and the quotient by it is semisimple. Each retained ℓ_i is therefore in the socle of $e_i(eAe)$. This indecomposable projective is injective and has simple socle; that socle is precisely its own simple head, represented by ℓ_i. This proves the claim. □

5.2 The corner algebra

For odd p, put $N = p-1$ and let Z_N be the zigzag algebra on vertices $0, \dots, N-1$. Opposite arrows join neighbors, non-returning paths of length two vanish, the two returns at an interior vertex agree, and all paths of length three vanish. Each arrow has degree one. The cubic relations are also imposed when $N = 2$. Denote its projectives by P_b and its endpoint simples by S_0, S_{N-1}. For $p = 2$, use instead

$$Z = \mathbf{k}[\epsilon]/(\epsilon^2), \quad |\epsilon|_{\text{int}} = 2,$$

with projective P and simple S.

Lemma 5.2. *In the free projective generator for $(\mathcal{C}_r)_{\Lambda_r}$, let e retain the old maximal-size blocks. For $r \geq 2$ its corner algebra is*

$$eB_{r+1}e \cong \begin{cases} Z \otimes B_r, & p = 2, \\ Z_{p-1} \otimes B_r, & p > 2, \end{cases}$$

where B_r has degree zero. Inflation of the old principal block is $X \mapsto S \boxtimes X$ or $S_0 \boxtimes X$.

Proof. For odd p the $p-1$ maximal blocks have simples

$$L^{[r]}_{\ell_b(j)}, \quad \ell_b(j) = \begin{cases} pj + b, & j \text{ even}, \\ pj + p - 2 - b, & j \text{ odd}, \end{cases} \quad 0 \leq b \leq p-2, \quad 0 \leq j < (p-1)p^{r-2}.$$

Their equivalence is part of [3, Proposition 4.54]. More explicitly, tensoring with V_r and projecting to an adjacent regular block sends the displayed simple to the one with the same j in that block. The restricted first-digit Clebsch–Gordan rule gives exactly the two neighbors; a possible wall has last digit $p-1$ and is omitted. The two translation functors are exact and adjoint. Their units and counits are isomorphisms on simples, and hence on all finite-length objects. Choose their composites along the line to identify every regular block with the principal one.

For free projectives the matrix formula is

$$\mathrm{Hom}_{\Lambda_r}(A \otimes \Lambda_r, B \otimes \Lambda_r) = \mathrm{Hom}_{\mathcal{C}_r}(A, B \otimes \Lambda_r).$$

The free projectives form a projective generator. Each is indecomposable: its endomorphism algebra has a local degree-zero part and a nilpotent positive-degree ideal. It shows that the corner has degree-zero part $\prod_b B_r$, degree-one identity bimodules between neighbors, and degree-two part one usual B_r-bimodule $x_b B_r$ at each vertex. All higher degrees vanish. We verify the multiplication. Choose generators c_{bc} of the neighboring identity bimodules. A return is $c_{bc}c_{cb} = x_b z_{bc}$ with $z_{bc} \in Z(B_r)$. Each z_{bc} is a unit. Indeed, the center of a connected finite-dimensional basic algebra is local. A central nonunit is in its radical and kills a nonzero primitive socle element $\ell_j \in e_j B_r e_j$. If z_{bc} were such an element, $c_{bc}\ell_j$ would be a socle element of the corner projective at (b, j) with simple label (c, j). It is killed by the old radical, all non-returning arrows and all returns, and by the degree-two part. This contradicts that this projective has its own head as socle.

Here the required socle property follows from Lemma 5.1, applied to the basic algebra B_{r+1}. It is proved independently of the proposed corner factorization and requires no assumption that this algebra is symmetric.

Normalize returns along the line. Set x_0 equal to its return, then set x_1 equal to the opposite return on that edge and rescale the next arrow by a central unit to make its return at vertex one agree. Continue. This gives exactly the zigzag relations over B_r. The displayed

degree-zero, one and two bimodules show the resulting surjection $Z_N \otimes B_r \to eB_{r+1}e$ is an isomorphism. It fixes the chosen degree-zero algebras.

For $p = 2$, the only maximal block is the even principal block. Tensoring with V_r changes parity, so the corner has degree zero B_r, degree one zero, and degree two the usual B_r bimodule. The latter is central and has square zero, giving $Z \otimes B_r$ directly. Finally an inflated object has all positive-degree arrows zero and lies at the unit endpoint. This proves the last assertion as an equality of exact functors. □

Lemma 5.3 (Endpoint cohomology)**.** *For odd p, the endpoint ring of Z_{p-1} is*

$$\mathrm{Ext}^\bullet(S_0, S_0) = \mathbf{k}[v] \otimes \bigwedge(\eta), \quad (|v|, w(v)) = (2p-2, 2p), \quad (|\eta|, w(\eta)) = (2p-3, 2p-2).$$

For the characteristic-two dual numbers it is $\mathbf{k}[v]$ with $(|v|, w(v)) = (1, 2)$. At level two these give $H_2 = \mathcal{E}_1(p)$, with $v = y_1$ under the normalization of the standard period, and with the respective degrees just specified.

Proof. The zigzag endpoint injective coresolution repeats the list

$$P_0, \dots, P_{N-1}, P_{N-1}, \dots, P_0.$$

Each traversal is a length-N extension between the endpoints. Maps between distinct adjacent projectives are arrows; repeated endpoint terms are joined by the socle loop. Here is the exactness check. The head and socle of P_b are both S_b. Its middle layer is $S_{b-1} \oplus S_{b+1}$, with the nonexistent neighbor omitted at an endpoint. In a forward traversal the incoming image in an interior P_b is the length-two branch with head S_{b-1} and socle S_b. The quotient is the length-two module with head S_b and socle S_{b+1}; this is exactly the image of the next arrow. At the far endpoint the incoming image is $\operatorname{rad} P_{N-1}$. The repeated-endpoint map has kernel $\operatorname{rad} P_{N-1}$ and image $\operatorname{soc} P_{N-1}$. The reverse traversal is checked with the two neighbors interchanged. At the other endpoint the same calculation returns to S_0. At the initial term the kernel is $\operatorname{soc} P_0 = S_0$. This proves exactness of the displayed repeated coresolution, also when $N = 2$. Derived Hom from S_0 is nonzero exactly in columns $2Nj$ and $2Nj + 2N - 1$, and each such group is one-dimensional. All induced differentials are zero, because each arrow or socle-loop map kills the socle of its source projective. The grading can be checked without inferring weights from cohomological degrees. Start P_0 with generator in degree -2 so that its socle is in degree zero. Each adjacent arrow lowers the next generator degree by one, and the repeated-endpoint socle map lowers it by two. The last P_0 in the displayed period therefore has generator degree $-2N - 2$ and socle degree $-2N$. The next first P_0 has generator degree $-2N - 4$ and socle degree $-2N - 2$. Thus the last socle gives η of bidegree $(2N - 1, 2N)$, and the periodic shift gives v of bidegree $(2N, 2N + 2)$. Periodicity makes all their indicated v-multiples nonzero; $\eta^2 = 0$ since its target degree has zero self-Ext. The products commute, either by the periodic chain maps or by the unit cohomology interpretation at level two. The first principal block is this Brauer line, with the unit at an endpoint, by [3, §5.1]. Its grading can also be read directly from (2.1): in the semisimple category $\mathcal{C}_1$, the fusion graph for V_1 is the line on $L_0, \dots, L_{p-2}$. The free-module matrix has one scalar in degree zero and two at each vertex, and one neighboring scalar in degree one. Its return products normalize as in Lemma 5.2. Thus its arrows have precisely the degree-one grading used here and $L_0 = \mathbf{1}$ is an endpoint, including the truncated algebra at $p = 3$. For the dual numbers, repeat its one-step socle sequence; the generator has cohomological degree one and weight two. This is the principal block of Ver_4. □

6 Restriction of a relative period, including its maps

Retain the notation of Section 3. Write $E = eB_{r+1}e$ and $U = S_0 \boxtimes \mathbf{1}$ when $p > 2$, or $U = S \boxtimes \mathbf{1}$ when $p = 2$. We use the corner factorization which fixes its degree-zero old blocks.

Proposition 6.1. *For odd p, the restriction of the first new Koszul antiperiod is the following exact splice. For each $k = 0, \dots, q-1$, traverse the $N = p-1$ projectives of Z_N with coefficient M_k, forward for even k and backward for odd k. Between traversal k and $k+1$, insert*

$$P_{e_k} \boxtimes M_k \xrightarrow{\mathrm{head}\boxtimes\iota_k} S_{e_k} \boxtimes Q_k \xrightarrow{\mathrm{socle}\boxtimes\pi_k} P_{e_k} \boxtimes M_{k+1}, \quad e_k = \begin{cases} N-1, & k \text{ even}, \\ 0, & k \text{ odd}. \end{cases} \tag{6.1}$$

The second half is the reversed native dual splice. For $p = 2$, there is one projective per traversal, and the restricted period is

$$0 \to U \to P \boxtimes M_0 \to S \boxtimes Q_0 \to P \boxtimes M_1 \to \cdots \to S \boxtimes Q_{a-1} \to P \boxtimes M_a \to U \to 0, \tag{6.2}$$

with the same head/socle wall maps. These are identifications of graded exact complexes up to rescaling terms and endpoints by nonzero scalars.

Proof. We first identify the terms. For odd p, write a regular coefficient index as $i = pk + b$, $0 \le b \le p-2$. Its old block is

$$b_k = \begin{cases} b, & k \text{ even}, \\ p-2-b, & k \text{ odd}. \end{cases}$$

Translation to block zero takes X_i to M_k. This follows by projecting adjacent Koszul sequences to the neighboring regular block: their other term belongs to the opposite neighbor, or to a deleted wall. Iterating reaches the index s_k of M_k. The functor taking a regular object to the corner restriction of its free module is $X \mapsto P_{b_k} \boxtimes \mathcal{T}_{b_k,0}(X)$. On projective generators this is the free-module matrix formula; exterior-degree-zero morphisms are the corresponding old morphisms. Exactness and projective presentations extend the statement naturally to the coefficient X_i.

At a wall $i = pk + p - 1$, $0 \le k < q-1$, the ordinary Steinberg formula transports to

$$X_i = J_r(Y_k) \otimes T_{p-1}.$$

In its free module, exterior degrees zero and two are deleted. The degree-one component is $J_r(Y_k) \otimes T_p$. It lies in regular endpoint block $p-2$ for even k, and block zero for odd k. Translation to zero identifies it with $D_r(Y_k) = Q_k$. To check the latter assertion on maps, for an old even projective T_i the successive neighboring translations take $T_p \otimes J_r(T_i)$ to $T_{2p-2} \otimes J_r(T_i)$. They commute with the Frobenius coefficient on these generators. For odd coefficients the object is already in block zero. Projective presentations give the stated natural identification for Y_k. All positive-degree Z_N arrows on this wall term are zero: their targets have deleted exterior degree two, or degree three. The term is therefore $S_{e_k} \boxtimes Q_k$, with the exterior-degree shift one.

For $p = 2$, an even coefficient $X_{2k} = M_k$ gives $P \boxtimes M_k$. An odd coefficient is $X_{2k+1} = J_r(Y_k) \otimes V_r$ by the Steinberg formula. Only its degree-one component survives; it is $J_r(Y_k) \otimes T_2 = Q_k$. Thus it gives $S \boxtimes Q_k$.

We now determine the maps. Between neighboring regular terms in one traversal the Hom space is a neighboring projective Hom of Z_N tensored with $\mathrm{End}(M_k)$; it is one-dimensional by Lemma 2.3. The two wall Hom spaces are one-dimensional by (3.3), and have precisely the maps in (6.1). Every such differential is nonzero. If the outgoing map from a regular term

were zero, the incoming radical or socle map would have to cover its Z_N-head, contradicting exactness. At a wall its incoming image is $M_k \subsetneq Q_k$, so its outgoing map is nonzero. The same argument applies to (6.2). Consequently successive scalar rescalings identify all maps with those displayed.

This also identifies the extension represented by the complex. A traversal with coefficient M_k is the endpoint Z_N-extension tensored with M_k. Follow it by the old short extension (3.1), tensored with the endpoint simple. Their Yoneda concatenation replaces the endpoint quotient by head $\boxtimes\, \iota_k$. Concatenating the next traversal with coefficient M_{k+1} replaces its initial inclusion by socle $\boxtimes\, \pi_k$. These are precisely the two displayed wall maps. Iterating gives the claimed Yoneda splice, including its maps, rather than merely a list of isomorphic middle terms. For $p = 2$, replace each endpoint traversal by its one-step socle extension.

For odd p, each regular translation commutes with native duality: it is tensoring with the self-dual natural object and projection to a dual-stable block. Together with $D_r(Y)^* = D_r(Y^*)$, this identifies the terms of the second half with the reverse-dual splice. The dual one-dimensional Hom spaces give its maps. At the two endpoint splices, the map is the quotient to the augmentation simple followed by its socle inclusion; it has exterior degree two and is unique up to scalar. There is no inserted old wall at those two splices. This proves the asserted identification of the whole period. A finite line of one-dimensional map spaces leaves only scalar endpoint factors after rescaling its terms. □

7 Coefficient restriction and truncated connecting maps

This section identifies the full coefficient maps, including nilpotent coefficients. Fix the coinduction adjunctions and the graded corner factorization used above.

Lemma 7.1 (Restriction on a free coefficient)**.** *Let X range through the old principal block. Restriction gives a map from extensions to its new free module to extensions to $P_0 \boxtimes X$ (or $P \boxtimes X$ for $p = 2$). Under derived coinduction and Künneth, this is a nonzero scalar multiple of the identity on $\mathrm{Ext}^\bullet_{\mathcal{C}_r}(\mathbf{1}, X)$. The assertion holds in every cohomological degree.*

Proof. The free coefficient functor restricts naturally to $X \mapsto P_0 \boxtimes X$ by the projective-presentation argument in Proposition 6.1. Both extension functors are naturally the old functor $\mathrm{Ext}^\bullet(\mathbf{1}, -)$. On the source this is derived coinduction; on the target the projective P_0 is also injective and $\mathrm{Hom}(S_0, P_0) = \mathbf{k}$. Restriction is a morphism of cohomological delta functors in X, since both free coinduction and corner restriction are exact.

In degree zero it is a natural endomorphism of $\mathrm{Hom}(\mathbf{1}, -)$, hence a scalar by Yoneda's lemma and $\mathrm{End}(\mathbf{1}) = \mathbf{k}$. This scalar is nonzero: for $X = \mathbf{1}$, restriction of the monic socle map from the unit is nonzero. The extension to positive degrees is a dimension-shifting argument. Choose an old exact sequence $0 \to X \to I \to X' \to 0$ with I injective. Coinduction takes I to a new injective, and $P_0 \boxtimes I$ is injective over the tensor-product algebra. The natural long exact sequences for restriction identify the degree-one map with the map on the cokernel of $\mathrm{Hom}(\mathbf{1}, I) \to \mathrm{Hom}(\mathbf{1}, X')$. It is multiplication by the same scalar. For $n \geq 2$ they identify the map with the degree-$(n-1)$ map for X'. Induction proves the assertion in every degree. Equivalently this is uniqueness for universal cohomological delta functors. For the dual numbers use its injective projective P.

The chosen term identifications can change this only by another scalar. Before localization they satisfy

$$\mathrm{End}^0_{E\text{-gr}}(P_0 \boxtimes M_k) = \mathrm{End}(M_k) = \mathbf{k},$$

and the same formula holds for M_k^* and for P. The superscript here refers to internal degree, not Ext degree. The degree-two socle loop cannot occur because the old coefficient

is concentrated in new degree zero. Thus the normalizations are fixed at the module level, where the relevant endomorphism space is one-dimensional. This proves the assertion for the term identifications of Proposition 6.1. □

Proposition 7.2 (Direct localized ring comparison)**.** *Restriction to the corner gives, over the actual old ring,*

$$H_{r+1}[t^{-1}] \cong \begin{cases} R[v], & p = 2, \\ R[v] \otimes \bigwedge(\eta), & p > 2, \end{cases} \qquad R = H_r[t^{-1}]. \tag{7.1}$$

The old factor has new weight zero; v, η have the degrees and new weights of Lemma 5.3.

Proof. Put $L = p^r - 1$ and $f_{\mathrm{new}}(n) = n + \lfloor n/L \rfloor$. Let $K^\bullet$ be the new relative coresolution and compare its hypercohomology spectral sequence with that of $eK^\bullet$. The latter is still an exact coresolution of $e\mathbf{1} = U$. Precomposition by the inflated old class t acts on both spectral sequences, compatibly with restriction and their truncation exact couples. Its degree is even when $p > 2$, and all signs are trivial when $p = 2$. It is the usual old action on the source coefficient groups by derived coinduction and Lemma 2.1. On the corner, old inflation is $X \mapsto S_0 \boxtimes X$ (or $S \boxtimes X$), so the image of t is $1 \otimes t$. In particular it has new internal weight zero.

Use Proposition 6.1 to compute the corner columns. A wall column is, by Künneth,

$$\mathrm{Ext}^\bullet_Z(S_0, S_e) \otimes \mathrm{Ext}^\bullet_{B_r}(\mathbf{1}, Q_k),$$

or its old native dual; in characteristic two both endpoints are read as S. Its old factor is killed by t, by Lemma 3.2, so it vanishes after localization. A regular term $P_b \boxtimes M_k$ contributes only for $b = 0$, because P_b is injective and its socle is S_b. It then contributes precisely the old group $\mathrm{Ext}^\bullet_{B_r}(\mathbf{1}, M_k)$; the same holds for native duals. For $p = 2$, use P instead of P_0. These are precisely the nonzero source coefficient columns, since the other X_i belong to different old blocks. Restriction is a scalar isomorphism on every remaining column by Lemma 7.1.

Localization and convergence can be checked in each new internal weight. A regular column of index n is pure of weight $f_{\mathrm{new}}(n) \geq n$. In a wall column the simple Z-module has generator in degree $-f_{\mathrm{new}}(n) - 1$. Simple-module Ext weights over Z are nonnegative: a homogeneous minimal projective resolution of a degree-zero simple has generators in nonnegative degrees. All wall weights are consequently at least $f_{\mathrm{new}}(n) + 1 \geq n$.

A fixed new weight w therefore uses only columns $0 \leq n \leq w$, including both halves and all subsequent periods. This bound is uniform under multiplication by t, whose new weight is zero. Localization of graded $\mathbf{k}[t]$-modules is exact and commutes with this finite filtration and its successive quotients. It thus commutes with the spectral sequence and its abutment. Localization permits negative cohomological degrees, but leaves the filtration length at w finite. After the walls vanish there is one surviving column at each remaining weight. The columnwise isomorphisms therefore identify the actual localized unit cohomology groups in every weight and degree. Taking their direct sum gives the ordinary graded localization; no completion in the new internal weight is involved.

Restriction is induced by an exact functor and preserves Yoneda products. It consequently gives an algebra isomorphism. Finally the global corner unit cohomology is $H_r \otimes \mathrm{Ext}^\bullet_Z(S_0, S_0)$ by Künneth, with the old inclusion as its old tensor factor. Localizing it at $1 \otimes t$ and applying Lemma 5.3 gives (7.1) and its claimed compatibilities. □

Corollary 7.3. *The generator v of (7.1) can be normalized so that the actual new standard parameter satisfies*

$$y_r = tv^q. \tag{7.2}$$

This is an equality in the full localized ring.

Proof. For odd p, there are q length-N endpoint antiperiods and $q-1$ old short extensions in the first half. Its length is $qN+(q-1)=pq-1$. The second half gives their reversed duals. The $2q$ endpoint antiperiods compose to v^q, and the old extensions compose to the old full period t, up to scalar. External Yoneda products commute with the usual interchange signs; N is even, so no sign occurs when moving a traversal past an old extension. For $p=2$ the q one-step endpoint periods and $q-1$ old short extensions compose to $v^q t$; all signs are trivial. Proposition 6.1 therefore sends the actual upper period to $\lambda t v^q$ with $\lambda \in \mathbf{k}^*$. Rescale v by a q-th root of λ, which exists in $\mathbf{k}$. This gives (7.2). □

Lemma 7.4 (A prefix represents its coefficient class)**.** *Let $\mathbf{1} \to K^0 \to K^1 \to \cdots$ be an exact graded coresolution, with cycle $A_n \hookrightarrow K^n$. Let $\varepsilon_n : A_n \to \mathbf{1}[n]$ be the extension of its first n terms. For $\gamma \in \mathrm{Ext}^e(\mathbf{1}, A_n)$, the class $\varepsilon_n \gamma$ has a total-cocycle representative concentrated in column n, whose E_1 symbol is the image of γ in $\mathrm{Ext}^e(\mathbf{1}, K^n)$ in the hypercohomology spectral sequence. In a fixed internal weight with only that column remaining on a localized E_1 page, this symbol determines the localized class uniquely.*

Proof. Work in one internal weight, shifting all target terms by that weight; suppress this common shift in the notation. Take a graded projective resolution $P^\bullet \to \mathbf{1}$, with $P^i = 0$ for $i > 0$. The column filtration of the double complex

$$D^{n,b} = \mathrm{Hom}(P^{-b}, K^n), \qquad n, b \geq 0,$$

gives the hypercohomology spectral sequence. Represent γ by a cocycle in $\mathrm{Hom}(P^{-e}, A_n)$. Its image in $D^{n,e}$ is a vertical cocycle. Its horizontal differential is zero because $A_n \subseteq \ker(K^n \to K^{n+1})$. Thus it is an actual total cocycle supported in that column, with the asserted E_1 symbol.

The cycle inclusion also gives a chain map $A_n[-n] \to K^\bullet$. Under $K^\bullet \simeq \mathbf{1}$ this is exactly the connecting morphism $A_n \to \mathbf{1}[n]$ represented by the first n short exact sequences, namely ε_n. Composing with γ therefore gives the total cocycle just described. This identifies the actual extension class. If only one column survives at the chosen weight, the induced filtration there has one nonzero successive quotient, so its symbol determines the class uniquely. Exact localization gives the same assertion whenever the weight has finite filtration, as verified in Proposition 7.2. □

Proposition 7.5 (The coefficient maps)**.** *Use the Künneth identification (7.1). In odd characteristic the localized images of the fundamental coefficient maps in (4.1) are, up to nonzero scalars and the usual graded signs, as follows:*

family	k	*old connecting map*	*corner image of* h
first	$2l$	δ_{2l}	$v^l \delta_{2l}(h)$
first	$2l-1$	δ_{2l-1}	$v^{l-1}\eta\, \delta_{2l-1}(h)$
second	$2l$	δ_{d-2l}	$v^{q-l-1}\eta\, \delta_{d-2l}(h)$
second	$2l-1$	$\delta_{d-(2l-1)}$	$v^{q-l} \delta_{d-(2l-1)}(h)$.

(7.3)

The range is $0 \leq k \leq a$ in each family. More explicitly, the even rows have $0 \leq l \leq (q-1)/2$, and the odd rows have $1 \leq l \leq (q-1)/2$. For $p=2$ the coefficient at k has localized image

$$v^k \delta_k(h), \qquad 0 \leq k \leq a, \tag{7.4}$$

with the same scalar convention. In particular these formulas identify the images of the unlocalized coefficient groups inside the full localized ring, rather than just their ranks or their reductions modulo the nilradical.

Proof. We explain precisely how a coefficient class is represented. Number the traversals in the odd-characteristic restricted period by $\nu = 0, \dots, 2q-1$. A first-family coefficient M_k has $\nu = k$ and old cycle index $j = k$. A second-family coefficient M_k^* has

$$\nu = 2q - 1 - k, \qquad j = d - k.$$

There is no old wall when passing between the two halves. Thus the prefix has accumulated exactly the first j old short extensions of (3.4); their composite is the actual old connecting extension $\delta_j : C_j \to \mathbf{1}[j]$.

If ν is even, the principal term is the first projective $P_0 \boxtimes C_j$ of its traversal. Its cycle is $S_0 \boxtimes C_j$. If ν is odd, the principal term is the last projective $P_0 \boxtimes C_j$ of its backward traversal. Its cycle is $\mathrm{rad}(P_0) \boxtimes C_j$. In the first case use the identity on the simple cycle; in the second use the socle map $S_0 \hookrightarrow \mathrm{rad}(P_0)$. Write this map into the cycle as β in either case. Composing it with the cycle inclusion into the free term is the socle map $S_0 \hookrightarrow P_0$ tensored with 1_{C_j}.

For an old coefficient class $h : \mathbf{1} \to C_j[e]$, form $\gamma = \beta(1_{S_0} \boxtimes h)$. Its image in the free term is exactly the old class h under Künneth and coinduction. The prefix extension composed with γ is therefore a representative for that column, by Lemma 7.4.

It remains to compute this explicit prefix product. The part in the Z_N direction has degree

$$c = \begin{cases} N\nu, & \nu \text{ even}, \\ N(\nu+1) - 1, & \nu \text{ odd}. \end{cases}$$

Indeed, in an even traversal the principal term is its first one, whereas in an odd traversal there are $N-1$ preceding projectives within that traversal. Composing with β gives exactly the corresponding endpoint resolution class from Lemma 5.3. It is $v^{\nu/2}$ when ν is even, and $v^{(\nu-1)/2}\eta$ when ν is odd. The old part is $\delta_j h$. This factorization follows from the literal Yoneda splice of Proposition 6.1; external Yoneda composition has only the usual interchange signs. Substitution of the two values of ν, j gives the four rows of (7.3). The total prefix length is $c + j$: it is s_k in the first family and $2p^r - 3 - s_k$ in the second, verifying the cohomological shifts as well.

Finally compare with the class in the original new ring. Restriction maps its column coefficient to a nonzero scalar multiple of h by Lemma 7.1. The explicit prefix representative just constructed has that same symbol after a scalar adjustment. After inverting t, all wall columns vanish and this new weight has only one remaining column. Lemma 7.4 therefore identifies the two actual classes in the localized ring. This proves the formulas for every unlocalized h. They extend to all localized coefficient classes by multiplying by powers of t; the maps commute with its actual Yoneda action. No degree-zero unit is substituted for this scalar comparison.

For $p = 2$, the regular term $P \boxtimes M_k$ has column index $2k$, old cycle index $j = k$, and cycle $S \boxtimes M_k$. Its prefix contains k endpoint periods and k old short extensions. The same socle lift has prefix product $v^k \delta_k h$, of degree $k + k = 2k$ before the coefficient degree is added. The identical one-column comparison gives (7.4). □

8 Exact descent and completion of the induction

Continue to assume the old identification $H_r = \mathcal{E}_{r-1}(p)$, including its internal grading. Let J_l be the span of its integral monomials whose x_{r-1} exponent is at least l. It is an old homogeneous ideal; exterior exponents are retained without change.

For odd p, x_{r-1} has old weight two, and (3.6) identifies the images in the four rows of (7.3) with

$$J_l, \quad J_l, \quad J_{q-l}, \quad J_{q-l},$$

respectively. To check the endpoints, $f(a) = q$ rather than $a = q - 1$, but weight $q - 1$ is absent by Remark 3.3. The case $j = d$ has tail $I_T = tH_r = J_q$. All other thresholds follow by rounding $f(j)/2$ up to the next integer, since old model weights are even. Explicitly, in this range

$$f(j) = \begin{cases} j, & 0 \le j < a, \\ q, & j = a, \\ j+1, & a < j < 2a, \\ 2q, & j = 2a. \end{cases}$$

For the first even row at $2l = a$, this gives $I_q = J_{l+1} = J_l$: the last equality is exactly the absence of weight $2l = q - 1$. For the second even row at $2l = a$, it gives $I_q = J_{q-l}$ directly. These are the only midpoint boundary checks in the four rows. The first rows cover the lower halves of the ranges below, and the second rows cover their upper halves. Hence the full fundamental image is

$$\bigoplus_{l=0}^{q-1} J_l v^l \ \oplus \ \bigoplus_{l=1}^{q} J_l v^{l-1} \eta. \tag{8.1}$$

For $p = 2$, x_{r-1} has weight one, and $I_{f(k)} = J_k$ for $0 \le k \le a$; at $k = a$ the skipped weight $q - 1$ gives $I_q = J_q = J_{q-1}$. The fundamental image is

$$\bigoplus_{k=0}^{q-1} J_k v^k. \tag{8.2}$$

By Proposition 4.1, the map from H_{r+1} to (7.1) is injective. Equations (4.1), (7.2), and (8.1)–(8.2) give its exact image: it is the indicated fundamental submodule and all its multiples by the actual parameter $y_r = tv^q$.

Lemma 8.1 (The model has the same image)**.** *Inside the common localized algebra* (7.1)*, the model $\mathcal{E}_r(p)$ has exactly that submodule as its image. Its old algebra and all its parameters agree with the actual ones.*

Proof. For odd p realize the new corner variables in the rationally graded ambient polynomial-exterior algebra by

$$v = x_{r-1}^{-1} x_r^p, \qquad \eta = x_{r-1}^{-1} x_r^{p-1} \xi_r. \tag{8.3}$$

They have integral degrees $2p - 2, 2p - 3$ and new weights $2p, 2p - 2$. An integral monomial has last polynomial exponent b and last exterior exponent $\varepsilon \in \{0, 1\}$ satisfying $p \mid b + \varepsilon$. This follows by clearing the denominators in $\sum_i (b_i + \varepsilon_i)/p^i \in \mathbb{Z}$. Indeed, integrality of the displayed rational degree is equivalent to this condition on the sum because p is odd. Multiplication by p^r then makes every term except the last divisible by p. Its last factor is therefore supplied by $v^{l-\varepsilon}\eta^\varepsilon$ with $pl = b + \varepsilon$. The remaining old factor is made integral by multiplying it by x_{r-1}^l. This proves the localized model identity

$$\mathcal{E}_r(p)[t^{-1}] = \mathcal{E}_{r-1}(p)[t^{-1}][v] \otimes \bigwedge(\eta).$$

It also gives exactly $x_r^{p^r} = x_{r-1}^{p^{r-1}} v^{p^{r-1}} = tv^q$.

For clarity, if A denotes the old ambient monomial, the factorization just used is

$$A x_r^{pl-\varepsilon} \xi_r^\varepsilon = (A x_{r-1}^l) v^{l-\varepsilon} \eta^\varepsilon.$$

The old coefficient on the right has integrality condition

$$\sum_{i<r} \frac{b_i + \varepsilon_i}{p^i} + \frac{l}{p^{r-1}} \in \mathbb{Z},$$

which is precisely the integrality condition on the left. Thus no projection to a reduced ring, or loss of old exterior factors, occurs in this coordinate change.

For the unlocalized image reduce $0 \le b < p^r$ by this new parameter. If $\varepsilon = 0$, one has $b = pl$ with $0 \le l < q$; if $\varepsilon = 1$, one has $b = pl - 1$ with $1 \le l \le q$. The remaining old monomial, after multiplication by x_{r-1}^l, has last exponent at least l, and is precisely an element of J_l. Conversely every monomial of $J_l v^l$ or $J_l v^{l-1}\eta$ has nonnegative exponents and integral degree in the new model. Thus the fundamental image is (8.1), including every choice of old exterior factors.

For $p = 2$ put $v = x_{r-1}^{-1} x_r^2$. Its degree is one and its new weight is two. Integrality forces the last exponent to be even. Reducing it modulo $2^r = 2q$ writes it as $2k$ with $0 \le k < q$. Multiplying the old part by x_{r-1}^k produces exactly a monomial of J_k. The converse is the same nonnegative-exponent check. This gives (8.2) and again $x_r^{2^r} = tv^q$. All older variables are unchanged. □

Proof of Theorem 1.1. The cases $m = 0, 1$ are semisimplicity of $\mathcal{C}_1$ and Lemma 5.3. Assume the theorem through H_r. The construction in Sections 3–4 proves freeness of H_{r+1} before localizing. Proposition 7.2 gives a ring localization over the actual old cohomology. The period and coefficient comparisons then give its exact image as the submodule described in Section 8. Lemma 8.1 identifies this with precisely the image of $\mathcal{E}_r(p)$ in the same localized ring. Both images are subalgebras. Their equality as submodules therefore identifies every product, including products of nilpotent classes, and descends to a graded algebra isomorphism $H_{r+1} = \mathcal{E}_r(p)$. It fixes the old algebra, preserves the actual new parameter by (7.2), and has the claimed last internal grading by (8.3) and its characteristic-two counterpart. This completes the compatible induction at every level.

For the rank, reduce each polynomial exponent b_i modulo p^i. For a fixed exterior subset in odd characteristic, the integral-degree condition is the single congruence

$$\sum_{i=1}^{m} p^{m-i}(b_i + \varepsilon_i) \equiv 0 \pmod{p^m}, \qquad 0 \le b_i < p^i.$$

The coefficient of b_m is one, so precisely $p^{m(m-1)/2}$ reduced exponent vectors satisfy it. There are 2^m exterior subsets. For characteristic two there are no exterior choices and the identical count is $2^{m(m-1)/2}$. The reduced monomials are a free basis over the specified polynomial parameters, proving the ranks. □

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